

EPHEMERIS

A generator of Berlin School music

Version 0.1.0 · 316 parameters · manual built 25.09.2026

The interface is divided into several sections:

- Top Bar:** Includes the title "EPHEMERIS", a "Cosmic" dropdown, a "A" dropdown, an "Aeolian" dropdown, and sliders for "Piece min" (set to 16.0) and "Concert min" (set to 0.00). Buttons for "Compose", "New seed", and "Stop" are present.
- Control Row:** A series of buttons for "reroll form", "reroll tempo", "reroll rows", "reroll rack", "reroll layers", "reroll lead", "reroll pads", "reroll hands", "Muted (env)", "Save set", "Load set", and "Export WAV + MIDI".
- Timeline:** Displays "seed 4242" and "16.0 min". A horizontal bar shows the progression of "nothing rerolled" with segments for "Einsatz", "Aufbau", "Lead", "Hoehepunkt", and "Ausklang".
- Parameter Section:** A grid of controls for various musical elements, each with a "Pan" knob, an "Echo" knob, and a "Hall" knob. The elements are organized into rows and columns:
 - Rows 1-8:** Each row has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knobs are set to -12.0, -11.0, -inf, -inf, -inf, -inf, -inf, and -inf respectively.
 - Lead:** Has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knob is set to -25.9.
 - Drone:** Has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knob is set to -inf.
 - Tape Keys:** Has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knob is set to -inf.
 - Strings:** Has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knob is set to -inf.
 - Drums:** Has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knob is set to -inf.
 - Atmosphere:** Has a "Pan" knob, an "Echo" knob, and a "Hall" knob. The "Hall" knob is set to -54.3.

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About

Ephemeris composes and plays long pieces in the manner of the modern Berlin School: sequencer lines of different lengths turning against each other over a common root, a bass that holds the ground, tape choirs and string machines, a lead that sings over the peak, the room of a tape echo and a long hall. There are no samples. Every sound is synthesised by the engine, and every note is decided by the composer.

The same seed always gives the same piece. A piece is a seed, a style, a length and a handful of rerolls, which is what a set file stores and what you can hand to someone else. The standalone and the plugin play what the offline renderer plays, sample for sample.

An ephemeris is the table that gives the positions of the heavenly bodies for every date. A sequencer is such a table, and the music here is made of orbits of different lengths around one root that meet again only now and then. The Rack page draws them that way.

The top bar

Style, key and scale choose the piece's character: Cosmic, Doom, Melodic, Modern and Drift are five style profiles with their own tempi, forms, layers and spaces. Piece min is the length of one piece; Concert min above zero strings pieces together into a concert that travels through related keys. With Morph To (on the Master page) a concert also wanders from its style to another: each piece is composed between the two, the further into the concert the nearer the second. Compose writes a piece with the current seed, New seed draws another one, Play starts and stops the standalone (in a host the host's transport plays).

The reroll buttons draw one part of the piece again and leave everything else as it was, to the note: the form, the tempo, the rows' shapes, the rack's patterns, the layers, the lead, the chords and pads, and the hands on the knobs. Save set and Load set keep a piece as a small text file; Export writes the piece offline as a 24-bit WAV and a MIDI file beside it, with the gestures as controllers -- and, if you choose so, the stems: a WAV per channel strip and one for the rooms in a folder beside the mix, whose sum is the mix before the master.

Below, the arrange view shows the sections of the piece -- the atmosphere at the start, the entry of the rows, the builds, the lead, the peak, the bridges, the fade -- with the playhead. Click to jump.

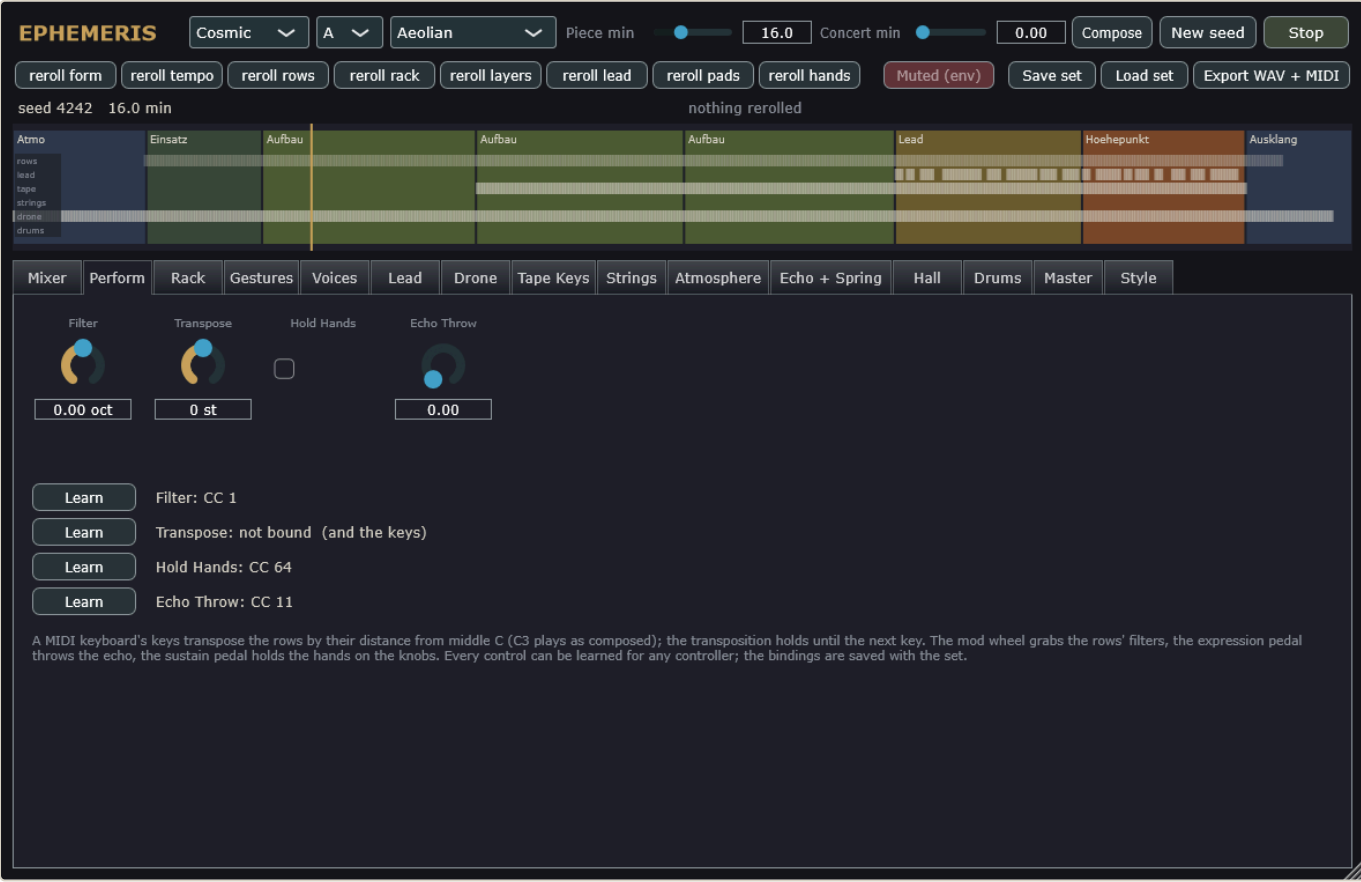
Mixer



A channel strip per source, in the order the engine mixes them: the eight rows, the lead, the drone, the tape keys, the string machine, the drums and the atmosphere. Each strip has its pan (where the source has one), its sends to the tape echo and to the hall, its fader and its meter. The controls are the same parameters the sources' own pages show.

The meter reads what the source puts into the mix after its fader and pan: a green bar for the RMS level with a release of 300 ms, turning amber above -6 dBFS and red above 0, and a peak line that holds for a second and a half before it falls at 20 dB per second. The figure under the meter is the held peak. A strip that shows -inf is not playing at the moment.

Perform



The performer's controls act on the piece as it plays, and at their defaults they do nothing at all: the piece is then exactly as composed. Filter moves the rows' filters up to two octaves either way, as a hand on the cutoff knobs of a modular would. Transpose is the transposition key: every note that starts after a change comes in transposed; notes that sound keep their pitch. Hold Hands lets go of the knobs: the gestures the composer wrote stand where they are until you release it. Echo Throw sends every source into the tape echo and raises the echo's feedback with it, for a dub throw at the end of a phrase.

A MIDI keyboard plays the transposition key directly: a key transposes by its distance from middle C and holds until the next key, as the transposer of a sequencer did. The mod wheel grabs the filters, the expression pedal throws the echo, the sustain pedal holds the hands. Learn binds any controller to any of the four; the bindings are saved with the plugin's state.

A row cannot be mutated live: the piece is composed before it plays. To hear the rack with other patterns, reroll the rack.

PERFORM

Control	Key	Range	Default
Filter	perform.filter	-2 .. 2 oct	0 oct
Transpose	perform.transpose	-12 .. 12 st	0 st
Hold Hands	perform.hold	off / on	off
Echo Throw	perform.throw	0 .. 1	0

Rack



On the left, the orrery. Every row that plays notes is a planet on an orbit, the bass innermost; one turn is one cycle of the row, its length in steps times its step. The top of every orbit is the row's first step. When running rows stand there together, a line of light rises from the centre: a conjunction, the moment the patterns come round together. Under the orbits, each row's cycle and the time until all running rows meet again. The sun in the middle is the transposer: its letter is the root the rows play on at the moment, and it moves when the transposer does.

The composer uses the conjunctions: a build, which brings in a new counter row, ends where that row and the bass row begin their cycles together again, if such a bar is near.

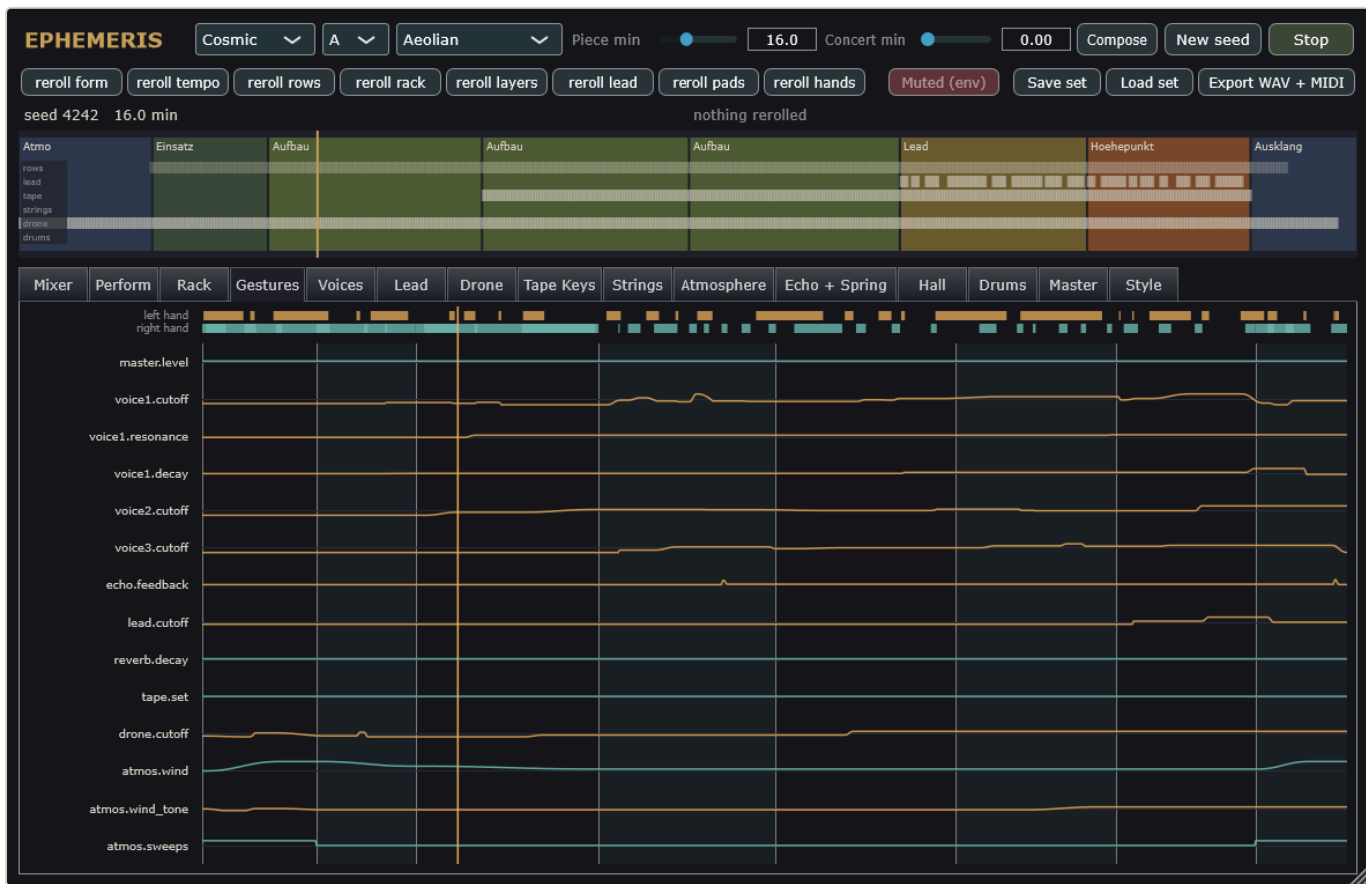
On the right, the rows' parameters: length and step, direction, octave and transposition, the chance of a mutation at the end of each cycle (a Turing machine's shift register), the gate, level, pan and sends, and the mode -- notes, or the transposer that moves the others. The composer sets the rows for each piece; the values here are what a piece starts from.

ROW (8 INSTANCES)

Control	Key	Range	Default
Active	rowN.active	off / on	on
Length	rowN.length	1 .. 32 steps	16 steps
Division	rowN.division	1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1 Bar, 2 Bars, 4 Bars	1/16
Direction	rowN.direction	Forward, Backward, Pendulum, Random Walk	Forward

Octave	rowN.octave	-3 .. 3	-1
Transpose	rowN.transpose	-12 .. 12 st	0 st
Mutation	rowN.mutation	0 .. 1	0
Gate	rowN.gate	5 .. 100 %	50 %
Level	rowN.level	-60 .. 6 dB	-6 dB
Pan	rowN.pan	-1 .. 1	0
Echo Send	rowN.echo	0 .. 1	0.3
Mode	rowN.mode	Notes, Transposer	Notes
Reverb Send	rowN.reverb	0 .. 1	0.3

Gestures



What the player's two hands do to the knobs over the whole piece. Every knob the composer moves has a lane: its offset from where you set the knob, as a curve over the piece, from -1 to +1 of the knob's range -- the filter of the bass row opening towards the peak and closing in the coda, the counter rows' filters, the lead's, the echo's feedback thrown now and then, the wind rising and falling in the atmosphere. The colour is the hand that moves the knob.

Above the lanes, a bar per hand shows where it is busy. A player has two hands, and the composer keeps to that: never more than two knobs move at once, and a hand finishes one move before it starts the next. The sections of the piece lie behind the lanes, the playhead runs over them, and a click jumps there. The hands can be let go on the Perform page (Hold Hands): every knob then stays where the curve has it.

Voices



Each row plays its own modular voice: two drifting oscillators (a sawtooth blended into a pulse, the second detuned), a mixer driven into a soft saturation, a four-pole transistor ladder at twice the sample rate, two envelopes and glide. The oscillators run freely and are never reset on a note, and each wanders slowly in pitch, as analogue oscillators do; the Drift knob sets how far.

The knobs the player's hands move during a piece -- cutoff, resonance, envelope amount, decay -- are moved by the composer's gestures as offsets from where you set them, so the ranges you choose keep their meaning.

VOICE (8 INSTANCES)

Control	Key	Range	Default
Wave	voiceN.wave	0 .. 1	0
Detune	voiceN.detune	0 .. 30 ct	7 ct
Pulse Width	voiceN.pw	0.05 .. 0.5	0.5
Drift	voiceN.drift	0 .. 15 ct	3 ct
Drive	voiceN.drive	0 .. 24 dB	6 dB
Cutoff	voiceN.cutoff	30 .. 16000 Hz	650 Hz
Resonance	voiceN.resonance	0 .. 1	0.4
Env Amount	voiceN.env_amount	0 .. 6 oct	3.5 oct
Filter Decay	voiceN.decay	15 .. 2000 ms	190 ms
Key Track	voiceN.keytrack	0 .. 1	0.5

Accent	voiceN.accent	0 .. 1	0.5
Amp Release	voiceN.amp_decay	5 .. 2000 ms	60 ms
Glide	voiceN.glide	1 .. 500 ms	60 ms

Lead



The lead is the same voice as a row's, with vibrato that comes in over the first half second of a held note, as a hand reaches the wheel. It plays in its own section and over the peak: phrases on the pentatonic of the moment, chord tones on the strong beats, glides from the tone below, and repeats of its own phrases.

LEAD

Control	Key	Range	Default
Wave	lead.wave	0 .. 1	0.25
Detune	lead.detune	0 .. 30 ct	4 ct
Pulse Width	lead.pw	0.05 .. 0.5	0.45
Drift	lead.drift	0 .. 15 ct	2 ct
Drive	lead.drive	0 .. 24 dB	8 dB
Cutoff	lead.cutoff	30 .. 16000 Hz	1500 Hz
Resonance	lead.resonance	0 .. 1	0.25
Env Amount	lead.env_amount	0 .. 6 oct	2 oct
Filter Decay	lead.decay	15 .. 2000 ms	450 ms
Key Track	lead.keytrack	0 .. 1	0.6
Accent	lead.accent	0 .. 1	0.3
Amp Release	lead.amp_decay	5 .. 2000 ms	300 ms
Glide	lead.glide	1 .. 500 ms	80 ms

Level	lead.level	-60 .. 6 dB	-9 dB
Pan	lead.pan	-1 .. 1	0.15
Echo Send	lead.echo	0 .. 1	0.45
Vibrato	lead.vibrato	0 .. 100 ct	18 ct
Vibrato Rate	lead.vibrato_rate	1 .. 10 Hz	5.2 Hz
Reverb Send	lead.reverb	0 .. 1	0.4

Drone



The drone is the ground under the whole piece: the root, low, held from the start to the fading coda, following the key when a phase changes it. It is a modular voice as well and has the lead's table.

DRONE

Control	Key	Range	Default
Wave	drone.wave	0 .. 1	0
Detune	drone.detune	0 .. 30 ct	9 ct
Pulse Width	drone.pw	0.05 .. 0.5	0.5
Drift	drone.drift	0 .. 15 ct	4 ct
Drive	drone.drive	0 .. 24 dB	4 dB
Cutoff	drone.cutoff	30 .. 16000 Hz	320 Hz
Resonance	drone.resonance	0 .. 1	0.3
Env Amount	drone.env_amount	0 .. 6 oct	0.5 oct
Filter Decay	drone.decay	15 .. 2000 ms	900 ms
Key Track	drone.keytrack	0 .. 1	0.3
Accent	drone.accent	0 .. 1	0
Amp Release	drone.amp_decay	5 .. 2000 ms	2000 ms
Glide	drone.glide	1 .. 500 ms	300 ms
Level	drone.level	-60 .. 6 dB	-10 dB

Pan	drone.pan	-1 .. 1	0
Echo Send	drone.echo	0 .. 1	0.1
Vibrato	drone.vibrato	0 .. 100 ct	0 ct
Vibrato Rate	drone.vibrato_rate	1 .. 10 Hz	5 Hz
Reverb Send	drone.reverb	0 .. 1	0.55

Tape Keys



A tape keyboard, synthesised: every key has its own strip of tape, with its own small detune, level and onset lag, played by one capstan whose speed wanders (wow and flutter) and sags a little under many keys. A tape runs out after eight seconds, as the instrument's did. Three sets: a choir of several singers through the formants of a sung vowel, strings, and flute. Age puts more of the machine's character into the sound; the tape set can change between pieces.

TAPE

Control	Key	Range	Default
Tapes	tape.set	Choir, Strings, Flute	Choir
Vowel	tape.vowel	0 .. 1	0.2
Wow	tape.wow	0 .. 30 ct	6 ct
Flutter	tape.flutter	0 .. 10 ct	2 ct
Motor Load	tape.sag	0 .. 5 ct	1 ct
Tone	tape.tone	1500 .. 16000 Hz	7000 Hz
Age	tape.age	0 .. 1	0.5
Level	tape.level	-60 .. 6 dB	-8 dB
Pan	tape.pan	-1 .. 1	-0.1
Echo Send	tape.echo	0 .. 1	0.1
Reverb Send	tape.reverb	0 .. 1	0.5

Strings

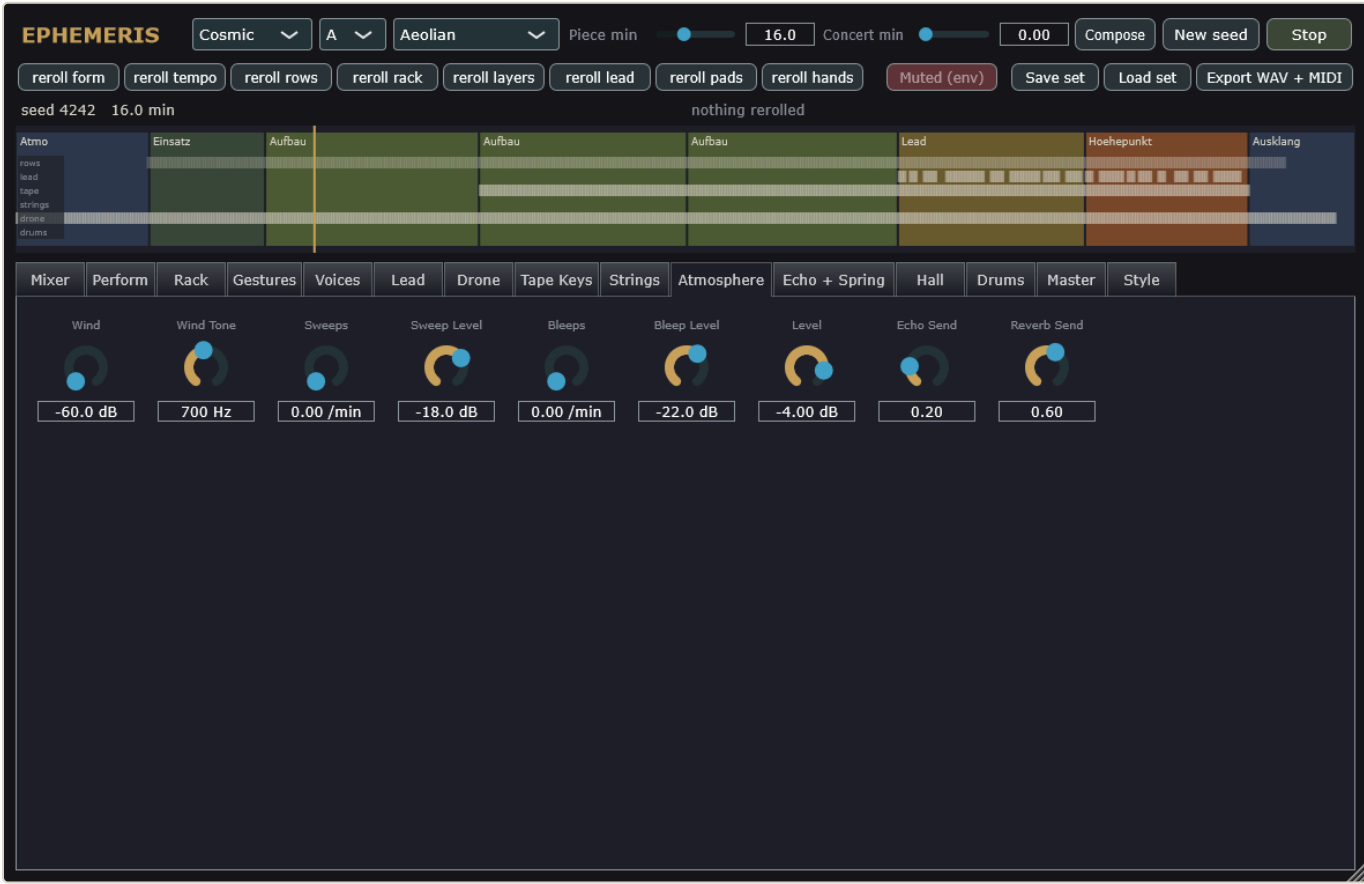


A string machine of the seventies: top-octave generators divided down into sawtooths at 8' and 4', through an ensemble of three bucket-brigade delays swept by a slow and a fast modulation. Keys glide from where they were when a chord is stolen, as the machine's did.

STRINGS

Control	Key	Range	Default
Crescendo	strings.attack	0.005 .. 3 s	0.35 s
Sustain	strings.release	0.05 .. 6 s	1.2 s
4' Mix	strings.feet	0 .. 1	0.4
Tone	strings.tone	800 .. 12000 Hz	4500 Hz
Ensemble	strings.ensemble	0 .. 1	0.8
Level	strings.level	-60 .. 6 dB	-10 dB
Pan	strings.pan	-1 .. 1	0.1
Echo Send	strings.echo	0 .. 1	0.05
Reverb Send	strings.reverb	0 .. 1	0.45

Atmosphere



Wind (filtered noise in a moving band), slow sweeps, and bursts of sample-and-hold bleeps on the notes of the scale: the space at the start of a piece and in the bridges, and the stars over the builds.

ATMOS

Control	Key	Range	Default
Wind	<code>atmos.wind</code>	-60 .. 0 dB	-60 dB
Wind Tone	<code>atmos.wind_tone</code>	150 .. 4000 Hz	700 Hz
Sweeps	<code>atmos.sweeps</code>	0 .. 6 /min	0 /min
Sweep Level	<code>atmos.sweep_level</code>	-60 .. 0 dB	-18 dB
Bleeps	<code>atmos.bleeps</code>	0 .. 12 /min	0 /min
Bleep Level	<code>atmos.bleep_level</code>	-60 .. 0 dB	-22 dB
Level	<code>atmos.level</code>	-60 .. 6 dB	-4 dB
Echo Send	<code>atmos.echo</code>	0 .. 1	0.2
Reverb Send	<code>atmos.reverb</code>	0 .. 1	0.6

Echo + Spring



A tape echo with a moving read head -- wow, flutter, saturation and loss of highs in the feedback -- synchronised to the tempo, with a ping-pong option; its feedback can run into self-oscillation, which the echo throw uses. Springs are built into it, as in the tape echoes of the time: they take the echo's send and return beside it.

ECHO

Control	Key	Range	Default
Time	echo.time	1/16, 1/8, 3/16, 1/4, 3/8, 1/2	3/16
Feedback	echo.feedback	0 .. 1.1	0.45
Tone	echo.tone	800 .. 12000 Hz	3500 Hz
Wow	echo.wow	0 .. 4 ms	0.6 ms
Flutter	echo.flutter	0 .. 0.5 ms	0.06 ms
Tape Drive	echo.drive	0 .. 18 dB	4 dB
Ping-Pong	echo.pingpong	off / on	on
Return	echo.return	-60 .. 6 dB	-4 dB

SPRING

Control	Key	Range	Default
Decay	spring.decay	0.3 .. 8 s	2.2 s

Tone	spring.tone	1500 .. 9000 Hz	4500 Hz
Return	spring.return	-60 .. 12 dB	-6 dB

Hall

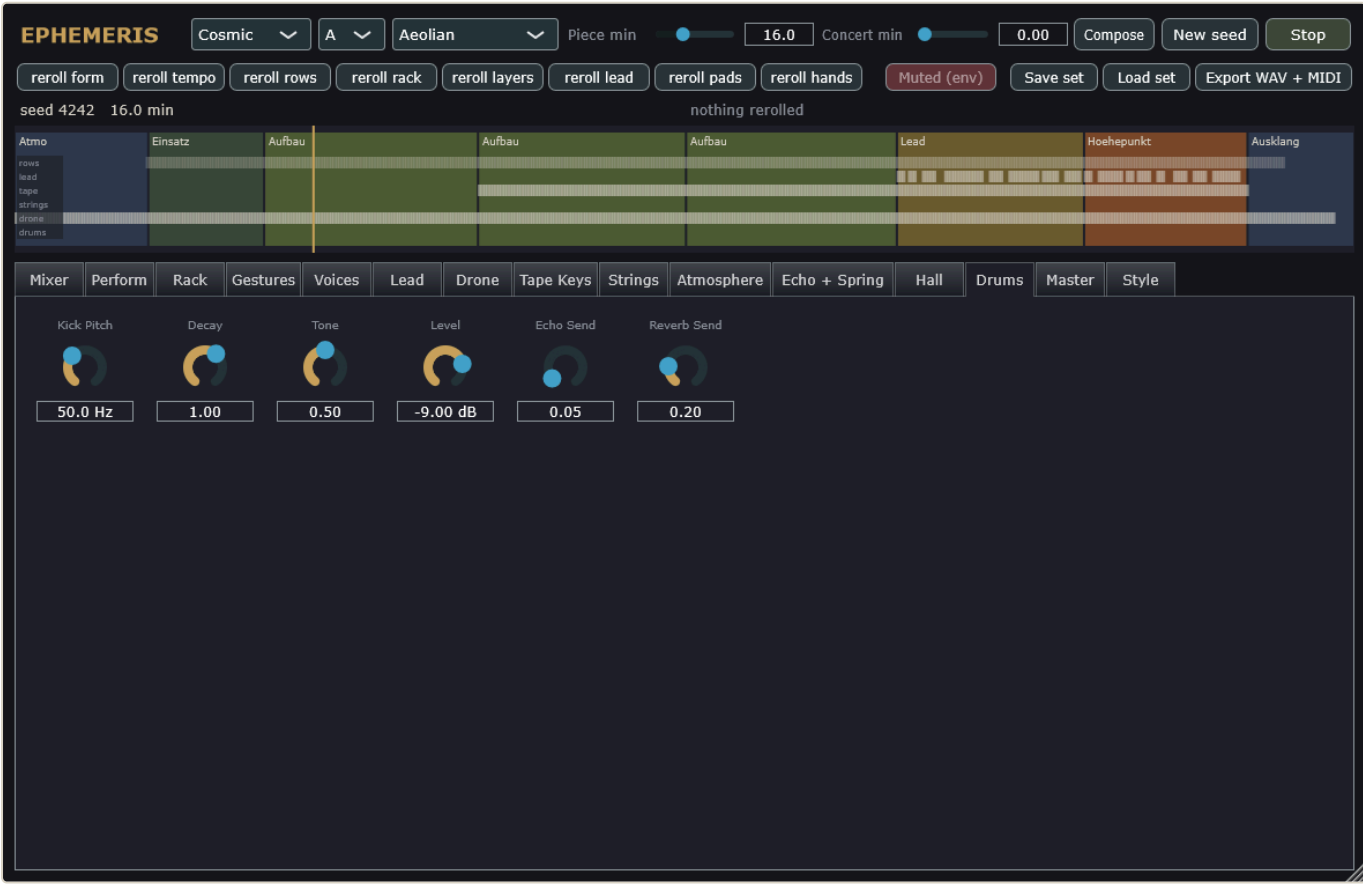


A long, dark hall (an eight-line feedback delay network) with its size, decay, damping, pre-delay and the band it returns.

REVERB

Control	Key	Range	Default
Size	reverb.size	0.3 .. 3	1.8
Decay	reverb.decay	0.3 .. 20 s	5.5 s
Damping	reverb.damping	0 .. 1	0.45
Pre-Delay	reverb.predelay	0 .. 200 ms	25 ms
Low Cut	reverb.lowcut	40 .. 500 Hz	110 Hz
High Cut	reverb.highcut	1000 .. 20000 Hz	9000 Hz
Return	reverb.return	-60 .. 12 dB	4 dB

Drums



A small analogue kit for the styles that have drums: kick, snare, hats, toms tuned to the root, rim and shaker, with patterns per style. The drums come late in a piece, from the second build to the end of the peak.

DRUMS

Control	Key	Range	Default
Kick Pitch	drums.kick_hz	35 .. 80 Hz	50 Hz
Decay	drums.decay	0.3 .. 2	1
Tone	drums.tone	0 .. 1	0.5
Level	drums.level	-60 .. 6 dB	-9 dB
Echo Send	drums.echo	0 .. 1	0.05
Reverb Send	drums.reverb	0 .. 1	0.2

Master



The master: its level, a gentle bus compressor that keeps the music's dynamics, and a true-peak limiter as protection. Below, the composer's settings -- the style, key, scale, tempo, lengths and whether the style chooses the tempo -- which the top bar also shows.

OSC Cues sends what the score knows to a visualiser such as Kaleidoscope, over UDP at the OSC Port, to the machine named by the environment variable EPH_CUE_HOST (this machine without it): every beat with the tempo (/eph/beat), every section with its phase (/eph/phase), every new root (/eph/key) and every conjunction of the rows with the share of the running rows that meet (/eph/conjunction). Each cue leaves when its sound is heard, not when it is composed.

MASTER

Control	Key	Range	Default
Level	master.level	-60 .. 6 dB	0 dB
Compress	master.compress	0 .. 1	0.5
Ceiling	master.ceiling	-6 .. 0 dBTP	-1 dBTP

COMPOSE

Control	Key	Range	Default
Tempo	compose.bpm	60 .. 160 BPM	118 BPM
Key	compose.key	C, C#, D, D#, E, F, F#, G, G#, A, A#, B	A

Scale	<code>compose.scale</code>	Aeolian, Dorian, Phrygian, Harmonic Minor, Minor Pentatonic	Aeolian
Style	<code>compose.style</code>	Cosmic, Doom, Melodic, Modern, Drift	Cosmic
Piece Length	<code>compose.piece_minutes</code>	4 .. 40 min	16 min
Style Tempo	<code>compose.style_tempo</code>	off / on	on
Concert Length	<code>compose.concert_minutes</code>	0 .. 240 min	0 min
Morph To	<code>compose.morph_to</code>	None, Cosmic, Doom, Melodic, Modern, Drift	None

CUE

Control	Key	Range	Default
OSC Cues	<code>cue.enabled</code>	off / on	off
OSC Port	<code>cue.port</code>	1024 .. 65535	9000

EPHEMERIS

Cosmic

A

Aeolian

Piece min

16.0

Concert min

0.00

Compose

New seed

Stop

reroll form

reroll tempo

reroll rows

reroll rack

reroll layers

reroll lead

reroll pads

reroll hands

Muted (env)

Save set

Load set

Export WAV + MIDI

seed 4242 16.0 min

nothing rerolled

Atmo

rows

lead

tape

strings

drums

Einsatz

Aufbau

Aufbau

Aufbau

Lead

Hoehepunkt

Ausklang

Mixer

Perform

Rack

Gestures

Voices

Lead

Drone

Tape Keys

Strings

Atmosphere

Echo + Spring

Hall

Drums

Master

Style

	Cosmic	Doom	Melodic	Modern	Drift
Tempo (BPM)	96 - 118	84 - 108	104 - 126	90 - 120	80 - 110
Piece in a concert (min)	12 - 22	10 - 18	7 - 12	6 - 11	12 - 28
Sequence phases	1 - 2	1 - 2	1 - 2	1 - 2	2 - 3
Atmosphere before / coda after	10 % / 8 %	12 % / 10 %	5 % / 5 %	8 % / 8 %	15 % / 12 %
New tempo / new key in a phase	30 % / 30 %	20 % / 40 %	40 % / 40 %	50 % / 50 %	70 % / 60 %
Rows at the peak	3	3	4	4	2
Counter row lengths	13 12 7 9	11 7 13	12 8 6 16	7 5 12 13	13 9
Mutation per cycle	20 %	15 %	10 %	30 %	30 %
Transposer	8 x 4 bars	6 x 4 bars	8 x 2 bars	8 x 2 bars	6 x 4 bars
Tape keys	90 %	70 %	60 %	50 %	80 %
String machine	50 %	30 %	60 %	70 %	40 %
Lead section	60 %	30 %	90 %	60 %	50 %
Bleeps	40 %	20 %	20 %	60 %	50 %
Drums	0 %	5 %	80 %	50 %	0 %
Tape set	Choir	Strings	Strings	Flute	Choir
Lead density	0.50	0.35	0.75	0.60	0.40
Hands: a move / a rest	8 s / 14 s	10 s / 16 s	7 s / 12 s	6 s / 11 s	12 s / 18 s
Darkness	-0.05	-0.15	0.05	0.00	-0.08
Hall	6.0 s	7.5 s	4.5 s	5.0 s	8.0 s
Level	4.0 dB	0.0 dB	3.0 dB	4.0 dB	0.0 dB

A style is a set of ranges and chances the composer draws from: two pieces of one style share these, not their notes. The tempo comes from the style while `compose.style_tempo` is on. The tempi and the levels were set after measuring reference recordings (`Tools/analyze_ref.py`).

The five style profiles side by side, the chosen one lit. A style is not a sound but a set of ranges and chances the composer draws from: the tempo, the length and the number of sequence phases, how much of the piece is atmosphere before the rows and after them, how often a later phase changes tempo or key, how many rows play at the peak and which lengths they are drawn from, how often a row mutates, how slowly the transposer moves, how often each layer comes in, how dense the lead is, how fast the hands move, how dark the voices are, how long the hall rings and how loud the piece is. Two pieces of one style share these, not their notes. The tempi and the levels were set after measuring reference recordings.